



Nguyen Quoc Viet

AI Researcher

📍 Seoul, South Korea | 📞 +82 10 2665 5066

✉ quocviet11102001@gmail.com | 🌐 visionquocviet

🐙 visiondrag | 🌐 visiondrag.github.io

Education

Sejong University

M.S. in Artificial Intelligence and Robotics

M.S. in Convergence Engineering for Intelligent Drone • Shared GPA: 4.29/4.5

Sep 2024–Aug 2026

Seoul, South Korea

VNU-HCM – University of Technology

B.E. in Computer Science • GPA: 8.55/10

Aug 2019–Nov 2023

Ho Chi Minh City, Vietnam

Academic Experience

VLILAB – Sejong University

Research Assistant

Sep 2024–Present

Seoul, South Korea

- Developed Key-Value (KV) Cache eviction method for efficient Computer-Use Agents (CUAs).
- Designed an efficient vision-language model (VLM) combining linear and quadratic attention for scalable long-context CUA.
- Developed a printed circuit board defect-detection pipeline using YOLO object detectors.

Sejong University

Teaching Assistant

Sep 2025–Jun 2026

Seoul, South Korea

- Supported undergraduate students on Machine Learning and Artificial Intelligence courses.

VNU-HCM – University of Technology

Teaching Assistant

Jun 2021–Jun 2022

Ho Chi Minh City, Vietnam

- Supported undergraduate students on Calculus, Physics, Linear Algebra, and Data Structures and Algorithms courses.

Publications & Patents

CUA-KV: A Two-Stage Context Eviction Method for Computer-Use Agents

Quoc-Viet Nguyen, Qiang Li, Cheol Jeong

Paper — Under Review

Jul 2026

- Proposes a two-stage context-eviction framework to reduce KV cache usage in long-horizon CUAs.
- Stage 1 performs temporal selection to retain only relevant historical interactions.
- Stage 2 performs spatial selection to preserve high-importance tokens in the KV cache.

Method and Apparatus for Long-Horizon Trajectory-Based Computer Agents

Cheol Jeong, Quoc-Viet Nguyen

Patent — Filed

Sep 2025

- Introduces a dual-branch transformer architecture that separates current-state reasoning from long-horizon interaction history.
- Uses linearized cross-attention and linear sequence modeling to compress historical trajectories into fixed-size representations.
- Reduces trajectory-processing complexity and prevents KV cache growth during extended computer-automation tasks.

Enhancing AI Chatbots for Mental Health Support: A Comprehensive Approach

Ngoc-Sang Vo, Ngoc-Thanh-Xuan Nguyen, Tan-Phuoc Pham, Quoc-Viet Nguyen, Hoang-Anh Pham

Paper — Published at MIWAI

Dec 2024

- Proposes a mental-health chatbot using retrieval-augmented generation (RAG) with vector-based and keyword-based retrieval to provide contextually relevant responses.
- Evaluates large language models (LLMs) for the application and selects Claude 3.5 Sonnet based on comparative performance.

RealCleanVSR: Mitigating the Perception-Distortion Trade-Off in Real-World Super-Resolution

Quoc-Viet Nguyen, Duc-Nam Tran, Minh-Vu Tran, Bao-Tam Bang-Ngoc

Paper — Preprint

Mar 2024

- Proposes two modules, Dyna-Mix Cleaning and Self-Reference Upsampling, to mitigate the perception-distortion trade-off in real-world video super-resolution.
- Demonstrates transferability of both modules to real-world image super-resolution by integrating them into Real-ESRGAN.

Industry Experience

CLT Vietnam

AI Engineer, Data Analyst

Jun 2022–Aug 2024

Ho Chi Minh City, Vietnam

- Developed video-processing algorithms for a factory recognition system.
- Investigated StyleGAN-based image-to-image translation for pet imagery.
- Prototyped 3D digital-avatar applications using Unity and Android ARCore.
- Led internal seminars and presentations on applied AI topics.
- Analyzed business reports and dashboards for Asia–Europe logistics operations.
- Investigated client-reported issues in TEU liftings, vessel utilization, and yield metrics.

Selected Projects

Multi-Stage Context Pruning for GUI Agents

Master Thesis

Mar 2026–Jun 2026

- Developed a multi-stage context-pruning method for GUI agents to remove redundant interaction history and reduce KV-cache usage in long-horizon tasks using Taylor approximation.
- Evaluated trade-offs among context length, memory usage, inference efficiency, and task performance on computer-use workflows.

Video Quality Enhancement Using Deep Learning-Based

Bachelor Thesis

Jan 2023–Jun 2023

- Studied real-world video super-resolution based on RealBasicVSR (CVPR 2022), with emphasis on recurrent propagation and artifact mitigation.
- Explored recurrent networks and generative adversarial training for perceptually convincing restoration.

Find It

Google Solution Challenge 2022

Mar 2022

- Built an Android indoor/outdoor navigation concept using Kotlin, Java, ARCore, Firebase, and Google Cloud Platform.
- Selected as a global Top 50 finalist in the 2022 Google Solution Challenge; the project addressed UN SDG 8: Decent Work and Economic Growth.

Technical Skills

Programming	Python (proficient), C/C++, MATLAB
AI Frameworks	PyTorch, TensorFlow, OpenCV, Hugging Face, ONNX, vLLM
Robotics	ROS 2, Gazebo
Generative AI	LLMs, VLMs, vision-language-action models (VLAs), AI agents
Cloud & Data	Google Cloud Platform, Firebase, Django, Looker
Collaboration	Git, GitHub, Jira, Slack, Google Workspace
Language	English — IELTS 7.5 (2023)

Honors, Awards & Certifications

BK21 Graduate Research Fellowship

Sejong University

Mar 2025–Mar 2026

Sejong University Scholarship

Sejong University

Sep 2024–Aug 2026

NASA International Space Apps Challenge

NASA

Oct 2025

Robotics in Space

EIT Deep Tech Talent Initiative

Sep 2025

LeRobot Worldwide Hackathon

Hugging Face

Jun 2025

17th Multi-Disciplinary International Conference on Artificial Intelligence

MIWAI

Dec 2024

VNU-HCMUT Academic Scholarship

VNU-HCMUT

Aug 2019–Nov 2023

Global Top 50 Finalist — Google Solution Challenge 2022

Google

Apr 2022

Research Interests

Deep learning architectures (transformers, diffusion models, attention mechanisms) • applied AI agents • embodied AI • VLAs for CUAs • VLAs for autonomous driving • planetary robotics